

A
PROJECT REPORT
ON
“AUTOMATED MOVIE TICKET BOOKING SYSTEM ”
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(CYBERSECURITY)

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CERTIFICATE

This is to certify that this Real Time Research Project Report entitled “**Automated Movie Ticket Booking System**” is a Bonafide work carried out by P. Kavya Jayasree 24N81A6231, A.Shruthi 24N81A6214, M.Pallavi 24N81A6229, L.Nikhitha 24N81A6228, J.Ashwini 24N81A6217 in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology in Computer Science and Engineering (CS)** from **Sphoorthy Engineering College**, affiliated to Jawaharlal Nehru Technological University Hyderabad, during the Academic Year 2025-26 under our guidance and supervision.

The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

We the undersigned, declare that the Real Time Research Project Report “**Automated Movie Ticket Booking System** ” carried out at “SPHOORTHY ENGINEERING COLLEGE” is original and is being submitted to the Department of COMPUTER SCIENCE AND ENGINEERING (CYBERSECURITY), Sphoorthy Engineering College, Hyderabad towards partial fulfilment for the award of degree of Bachelor of Technology.

We declare that the result embodied in the the Real Time Research Project Report has not been submitted to any other University or Institute for the award of any Degree or Diploma.

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ABSTRACT

The Automated Movie Ticket Booking System is a software application designed to simplify and automate the process of booking movie tickets. Traditional ticket booking methods often involve long queues, manual seat allocation, and time-consuming processes. This system aims to overcome these challenges by providing a user-friendly digital platform where customers can easily browse available movies, select show timings, choose seats, and complete the booking process efficiently.

The application is developed using programming technologies such as Python, Tkinter for the graphical user interface, and JSON for data storage. The system allows users to view movie details, check seat availability, select preferred seats, and make payments through multiple methods such as QR code, card, or cash. After successful payment, the system generates an e-ticket or receipt for the user.

One of the key features of this system is the automated booking scheduler, which manages opening and closing of bookings based on predefined show timings. This reduces manual intervention and improves the overall efficiency of ticket management. The system also maintains booking records and ensures proper seat allocation to avoid double bookings.

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CHAPTER-1

INTRODUCTION

The Automated Movie Ticket Booking System is a software application developed to simplify and improve the process of booking movie tickets in theatres. In traditional ticket booking systems, customers often need to stand in long queues at ticket counters to purchase tickets. This process can be time-consuming and may lead to inconvenience, especially during weekends, holidays, or when popular movies are released. Manual booking methods may also lead to errors in seat allocation and record management.

To overcome these problems, the Automated Movie Ticket Booking System provides a computerized platform where users can easily view available movies, check show timings, select seats, and book tickets in a convenient manner. The system is designed to make the booking process faster, more organized, and user-friendly. It reduces the workload of theatre staff by automating tasks such as seat management, ticket generation, and record maintenance.

The application is developed using Python programming language, with Tkinter for creating the graphical user interface (GUI) and JSON for storing data such as movie details and booking records. The system allows users to choose their preferred seats, make payments through available methods, and receive an electronic ticket after successful booking.

1.1 PROBLEM STATEMENT

In many cinema theatres, movie tickets are still booked through manual processes at ticket counters. This traditional method often leads to several problems such as long waiting lines, delays in ticket booking, and inconvenience for customers. During peak times like weekends, holidays, or the release of popular movies, managing large numbers of customers becomes difficult for theatre staff. Manual ticket booking systems may also cause errors in seat allocation, double booking of seats, and difficulties in maintaining proper booking records.

In addition, customers may not have access to real-time information about available seats or show timings, which can lead to confusion and dissatisfaction. Another major issue is the lack of an organized system to manage bookings efficiently. Theatre staff must manually handle ticket sales, seat availability, and payment records, which increases workload and reduces efficiency.

Therefore, there is a need for an Automated Movie Ticket Booking System that can streamline the ticket booking process, allow users to view movie details and seat availability, select seats easily, and generate tickets automatically. Such a system can reduce manual work, minimize errors, save time for customers, and improve the overall efficiency of theatre ticket management.

1.2 OBJECTIVE

The main objective of the Automated Movie Ticket Booking System is to develop a software application that simplifies and automates the process of booking movie tickets. The system aims to provide a convenient and user-friendly platform where customers can easily browse available movies, check show timings, select seats, and book tickets without standing in long queues at ticket counters. It also helps in preventing errors such as double booking of seats and ensures proper seat allocation.

The system automatically generates electronic tickets after successful booking and maintains booking records for future reference. Overall, the project aims to reduce waiting time for customers, decrease manual work for theatre staff, and improve the efficiency and organization of the movie ticket booking process.

1.3 MOTIVATION

The motivation behind developing the Automated Movie Ticket Booking System is to improve the traditional method of booking movie tickets and make the process faster, easier, and more efficient. In many theatres, customers still face problems such as long queues, waiting time, and manual errors during ticket booking. These issues can cause inconvenience for customers and increase the workload for theatre staff. With the growth of digital technology, there is a need for systems that can automate such processes and provide better services.

This project is motivated by the idea of using software technology to simplify ticket booking, manage seat allocation efficiently, and provide a convenient platform for users to book tickets. It also helps students understand how real-world systems can be designed and implemented using programming tools and technologies.

1.4 EXISTING SYSTEM

In the existing system, movie tickets are mostly booked through manual methods at theatre ticket counters. Customers need to visit the cinema theatre in person and stand in long queues to purchase tickets. The theatre staff manually checks seat availability and issues tickets to customers. This process is time-consuming and often leads to inconvenience, especially during weekends, holidays, or when popular movies are released. The manual booking system may also lead to problems such as human errors in seat allocation, double booking of seats, and difficulty in maintaining accurate records of bookings and payments.

In addition, customers may not have clear information about available seats, show timings, or ticket availability before reaching the theatre. Due to the lack of automation, theatre staff must manage ticket sales, seat allocation, and record keeping manually, which increases workload and reduces efficiency. Therefore, the existing system is not always reliable, fast, or convenient for both customers and theatre

management. These limitations highlight the need for an automated ticket booking system.

1.5 PROPOSED SYSTEM

The Proposed System is the Automated Movie Ticket Booking System, which is designed to overcome the limitations of the existing manual ticket booking process. This system provides a computerized platform where users can easily browse available movies, view show timings, select seats, and book tickets through a user-friendly interface. The system reduces the need for manual work and improves the overall efficiency of ticket management.

In this system, users can check seat availability in real time and choose their preferred seats before confirming the booking. The application also supports different payment methods such as QR code, card, or cash, making the payment process simple and flexible. After successful booking, the system automatically generates an electronic ticket or receipt for the user. The proposed system also maintains booking records and manages seat allocation to avoid issues such as double booking.

An automated booking scheduler can control when ticket booking opens or closes based on predefined show timings. Overall, the proposed system makes the movie ticket booking process faster, more accurate, and more convenient for both customers and theatre administrators.

1.6 SCOPE

The scope of the Automated Movie Ticket Booking System is to develop a simple and efficient software application that automates the process of booking movie tickets. The system allows users to view available movies, check show timings, and select their preferred seats through a user-friendly interface. It also enables customers to book tickets easily without the need to stand in long queues at theatre counters.

The system manages important functions such as seat availability, ticket booking, payment processing, and ticket generation. It maintains booking records and ensures that seats are allocated properly to avoid issues such as double booking. The application can be used by small cinema theatres or for educational purposes to demonstrate how ticket booking systems work.

The project mainly focuses on providing a convenient and organized platform for movie ticket reservations while reducing manual work and improving efficiency. In the future, the system can be expanded by adding features such as online payment gateways, mobile application support, and integration with larger theatre management systems.

1.7 SOFTWARE & HARDWARE REQUIREMENTS

SOFTWARE REQUIREMENTS

PROGRAMMING LANGUAGES:

The Automated Movie Ticket Booking System is developed using Python as the main programming language. Python is used to implement the core functions of the system such as ticket booking, seat selection, and record management. The system also uses Tkinter to create the graphical user interface (GUI), which allows users to interact with the application easily. JSON is used for storing data such as movie details, seat availability, and booking records. These technologies help in developing a simple and efficient ticket booking system.

LIBRARIES & PACKAGES:

The Automated Movie Ticket Booking System uses several libraries and packages to support its functionality. Tkinter is used to create the graphical user interface (GUI) for user interaction. JSON is used for storing and managing data such as movie details and booking records. The Datetime library is used to manage show timings and booking schedules. These libraries help in building an efficient and user-friendly system.

DEVELOPMENT ENVIRONMENT / IDE:

The Automated Movie Ticket Booking System is developed using PyCharm as the Integrated Development Environment (IDE). PyCharm provides useful features such as code editing, debugging, and project management, which make the development process easier and more efficient. It supports Python programming and helps in organizing the project files properly during development.

JDK REQUIREMENTS:

The Automated Movie Ticket Booking System does not require JDK (Java Development Kit) because the project is developed using Python programming language. Instead of JDK, the system requires Python to be installed on the computer to run the application. Python provides all the necessary tools and libraries needed for the development and execution of the system.

OPTIONAL TOOLS:

Some optional tools can be used to improve the development and functionality of the Automated Movie Ticket Booking System. GitHub can be used for version control and to store the project code online. QR Code Generator libraries can be used to generate QR codes for payment or e-tickets. Streamlit or similar tools can also be used if the project is later converted into a web-based application. These tools are not mandatory but can enhance the project features and management.

OPERATING SYSTEM:

The Automated Movie Ticket Booking System can run on operating systems such as Windows, Linux, and macOS. However, the project is mainly developed and tested on the Windows operating system, which provides a stable environment for running Python applications and supporting development tools like PyCharm.

HARDWARE REQUIREMENTS

PROCESSOR:

The system requires a basic processor such as Intel Core i3 or higher to run the Automated Movie Ticket Booking System efficiently. Since the application is developed using Python and Tkinter, it does not require a high-performance processor. Any standard processor capable of running Python and the development environment smoothly is sufficient.

RAM:

The system requires a minimum of 4 GB RAM to run the Automated Movie Ticket Booking System smoothly. This amount of memory is sufficient for running Python, the development environment (such as PyCharm), and the application without performance issues.

DISPLAY:

The system requires a standard monitor with a minimum resolution of 1024×768 pixels. A clear display is necessary to properly view the graphical user interface (GUI) of the application, including movie details, seat selection, and booking options.

INPUT DEVICES:

The system requires basic input devices such as a keyboard and mouse. The keyboard is used to enter information like user details and booking data, while the mouse is used to navigate through the graphical user interface and select options such as movies, seats, and booking buttons.

NETWORK:

The system does not require a network connection since it is a desktop-based application. However, an internet connection may be used for features like online payments or updates.

CHAPTER 2

LITERATURE SURVEY

The Automated Movie Ticket Booking System is developed based on the concept of online and computerized ticket reservation systems used in modern cinemas. Many theatres use digital platforms that allow customers to view movie schedules, select seats, and book tickets easily. These systems help reduce long queues at ticket counters and improve the overall efficiency of ticket management.

Previous studies and applications show that automated booking systems provide better convenience for users and reduce manual errors in seat allocation and record maintenance. They also help theatre management organize bookings and track ticket sales effectively.

This project is inspired by such systems and aims to develop a simplified desktop-based movie ticket booking application using programming technologies like Python and Tkinter. It demonstrates how automation can improve the traditional ticket booking process.

2.1 SURVEY OF MAJOR AREA RELEVANT TO PROJECT

The Automated Movie Ticket Booking System belongs to the broader field of online reservation systems, information management systems, and software-based service automation. These systems are widely used in industries such as cinema theatres, airlines, railways, hotels, and event management to simplify the process of booking services and managing customer data efficiently.

The main goal of these systems is to reduce manual effort, improve accuracy, and provide faster service to users. One of the major areas related to this project is Online Reservation Systems. These systems allow users to book tickets or services through computerized platforms. In the context of cinema theatres, online reservation systems enable customers to view available movies, check show timings, select seats, and confirm bookings without visiting the theatre physically.

These systems help in reducing long queues and provide convenience for customers. Another important area is Database Management Systems (DBMS). Ticket booking systems require proper storage and management of data such as movie details, seat availability, user information, and booking records. Efficient data storage ensures that the system can retrieve and update information quickly. In this project, JSON files are used for storing and managing such data in a structured format.

The project is also related to Graphical User Interface (GUI) development, which focuses on designing interactive and user-friendly interfaces. A well-designed GUI allows users to easily interact with the system, navigate through options, and perform tasks such as selecting movies or booking tickets. In this project, Tkinter is used to develop the GUI, providing buttons, menus, and windows for user interaction. Another major area relevant to this project is Software Automation. Automation helps

in performing tasks automatically without human intervention. In the Automated Movie Ticket Booking System, processes such as seat allocation, ticket generation, and booking record maintenance are automated. This reduces human errors and increases the efficiency of the system.

Additionally, the system relates to Payment Processing Systems, which handle the transaction process when users book tickets. Although this project mainly demonstrates simplified payment options, modern ticket booking systems often integrate secure online payment gateways to allow users to pay using cards, digital wallets, or QR codes. Overall, the major areas relevant to this project include reservation systems, database management, graphical user interface development, and software automation.

By combining these areas, the Automated Movie Ticket Booking System demonstrates how technology can be used to improve traditional ticket booking processes and provide a more efficient and convenient experience for users.

2.2 TECHNIQUES AND ALGORITHMS

The Automated Movie Ticket Booking System uses simple techniques to manage the booking process. A Graphical User Interface (GUI) using Tkinter allows users to interact with the system easily. JSON data handling is used to store movie details and booking records. A seat availability checking algorithm ensures that seats are not double-booked. These techniques help the system run efficiently and manage ticket bookings accurately.

Key Techniques and Methodologies

The Automated Movie Ticket Booking System uses several key techniques and methodologies to ensure efficient functioning. The system uses GUI-based interaction through Tkinter to provide an easy-to-use interface for users. Data management using JSON is used to store and retrieve information such as movie details and booking records. The system also applies seat availability checking to prevent double booking and ensure accurate seat allocation. Additionally, automation techniques are used to generate tickets and manage booking records efficiently.

Performance Considerations

The Automated Movie Ticket Booking System is designed to provide smooth and efficient performance. The system ensures fast response time when users browse movies, select seats, and confirm bookings. Efficient data handling using JSON files helps in quickly storing and retrieving booking information. The application also ensures accurate seat management to avoid conflicts such as double booking. Since the system is lightweight and desktop-based, it requires minimal system resources and can run smoothly on standard computers. These considerations help maintain reliability and a better user experience.

2.3 APPLICATIONS

Cinema Theatre Ticket Booking

The Automated Movie Ticket Booking System can be used in cinema theatres to automate the process of ticket reservations. Customers can view movie details, check show timings, select seats, and book tickets quickly without standing in long queues at ticket counters.

Multiplex Theatre Management

The system is useful for multiplex theatres that run multiple movies at different times. It helps theatre management organize seat allocation, manage show schedules, and maintain booking records efficiently.

Small Theatre and Local Cinema

Use Small cinema halls and local theatres can use this system to replace manual ticket booking methods. It helps them manage ticket reservations digitally and maintain proper records of bookings.

Educational and Academic Projects

The project can be used by students as a learning model to understand how real-world booking systems work. It helps in learning concepts such as GUI development, data management, and automation using programming.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The System Architecture of the Automated Movie Ticket Booking System describes how different components of the system interact with each other to perform the ticket booking process.

User Interface (GUI Layer)

This layer is responsible for user interaction with the system. It is developed using Tkinter and provides options for users to view movies, select seats, enter booking details, and confirm ticket reservations.

Application Logic Layer

This layer handles the main functionality of the system. It processes user requests such as checking seat availability, managing bookings, and generating tickets. The logic of the system is implemented using Python programming. .

Data Management Layer

This layer is responsible for storing and retrieving data. Information such as movie details, seat availability, and booking records is stored using JSON files. The system reads and updates this data whenever a booking is made.

Booking and Seat Management Module

This module manages seat selection and booking confirmation. It checks whether seats are available and prevents double booking by updating the seat status after each reservation.

Ticket Generation Module

After successful booking, this module automatically generates an electronic ticket or receipt for the user containing details such as movie name, show time, seat number, and booking confirmation.

Payment Handling Module

This module manages the payment process. It supports different payment methods such as QR code, card, or cash, and confirms the booking once payment is completed. Together, these components form the overall system architecture and ensure that the movie ticket booking process runs smoothly and efficiently.

Overall Architecture Benefits:

Better Organization

The system architecture separates the user interface, logic, and data management, making the system well-structured. .

Improved Efficiency

Each module performs a specific task, which helps the system work faster and more accurately. .

Easy Maintenance

The modular structure makes it easier to update or modify the system. .

Scalability

New features such as online payments or mobile access can be added in the future. .

Reliability

The system helps prevent errors like double booking and maintains accurate booking records.

3.2 SYSTEM FLOW

• Start the Application

The user opens the Automated Movie Ticket Booking System application.

• View Movies and Show Timings

The system displays the list of available movies along with their show timings.

• Select Movie and Seats

The user selects a movie and chooses the preferred seats from the available options.

• Check Seat Availability

The system verifies whether the selected seats are available.

• Confirm Booking and Payment

The user confirms the booking and completes the payment process.

• Ticket Generation

After successful booking, the system generates an electronic ticket with booking details.

• End Process

The booking information is stored in the system, and the process is completed.

3.2.1 Data flow Diagram Steps:

- **User Input**

User selects movie, show time, and seats in the system

- **Process Request**

The system processes the user request and checks seat availability.

- **Retrieve Data**

Movie details and seat information are retrieved from stored data (JSON).

- **Booking Confirmation**

If seats are available, the system proceeds with booking and payment.

- **Update Data**

Seat status and booking records are updated in the system.

- **Ticket Generation**

The system generates an electronic ticket

- **Output to User**

Booking confirmation and ticket details are displayed.

[Start]



[Open Application]



[Select Movie & Seats]



[Check Availability]



[Confirm Booking]



[Generate Ticket]



[End]

Fig 3.2.1 Data flow Diagram

3.2.2 Use case Diagram

Use case diagram is a graphic depiction of the interactions among the elements of a system. Use cases will specify the expected behaviour, and the exact method of making it happen. Use cases once specified can be denoted both textual and visual representation.

Use case diagrams are used to specify:

- Requirements (external), required usages of a system under design or analysis-to capture what the system is supposed to do.
- The functionality offered by a subject– what the system can do.
- Requirements the specified subject poses on its environment- by defining how environment should interact with the subject so that it will be able to perform its services.

i. Requirements (external):

Use case diagrams are used to capture the external requirements of a system. These requirements represent the expected usages of the system under design or analysis. They define what the system is supposed to do from the perspective of an external observer (like a user or another system). Each use case in the diagram represents a specific requirement or functionality that the system should provide.

ii. Functionality offered by a subject:

The subject in a use case diagram is typically the system under design or analysis. The use cases associated with the subject represent the functionality that the system offers. This could include various tasks or services that the system can perform. For example, in a banking system, the use cases might include “Withdraw Money”, “Deposit Money”, “Check Balance”, etc. Each of these use cases represents a specific functionality that the banking system offers.

iii. Requirements the specified subject poses on its environment:

Use case diagrams also capture the requirements that the system (subject) poses on its environment. This includes defining how the environment (users or other systems) should interact with the system so that it can perform its services. These interactions are represented as relationships between the use cases (system’s services) and the actors (users or other systems) in the diagram.

Representation:

Use case diagrams provide both a textual and a visual representation of the system’s requirements. The visual representation is a diagram that shows the use cases, actors, and their relationships. The textual representation provides more detailed descriptions of each use case, including the steps involved in the use case (also known as the use case scenario or flow of events), preconditions, postconditions, and any alternative or exception flows.

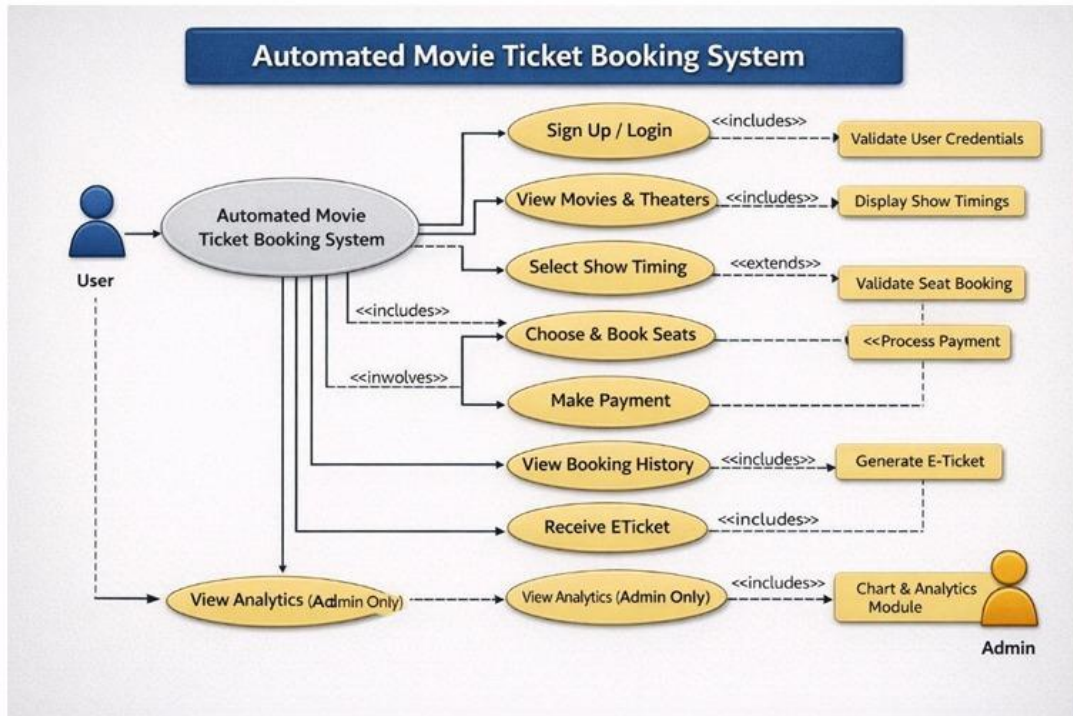


Fig 3.2.2 Use Case Diagram (Automated Movie Ticket Booking System)

Actors:

- **User**
- **Admin**

Use Cases:

1. User

- View Movies
- Select Show Time
- Choose Seats
- Make Payment
- Book Ticket
- Receive E-Ticket

2. Admin

- Add / Update Movie Details
- Manage Show Timings
- View Booking Records

3.2.3 Sequence Diagram

Steps in Sequence Diagram

Stepwise Actions for the Sequence Diagram

1. **User** → **Login System**: Enter username & password
2. **Login System** → **User Details JSON**: Validate credentials.
3. **Login System** → **User**: Login success/failure.
4. **User** → **Main Window**: Select a movie.
5. **Main Window** → **Theater & Show Data**: Display theaters, dates, and show times.
6. **User** → **Main Window**: Select date, time, and seats.
7. **Main Window** → **Bookings JSON**: Check seat availability.
8. **Main Window** → **Payment Module**: Select payment method.
9. **Payment Module** → **Payment Gateway / QR**: Process payment.
10. **Payment Module** → **Bookings JSON**: Update booked seats.
11. **Payment Module** → **User Bookings JSON**: Update user history.
12. **Main Window** → **User**: Show ticket confirmation.
13. **Analytics Module (Optional)**: Read bookings JSON → Display charts.

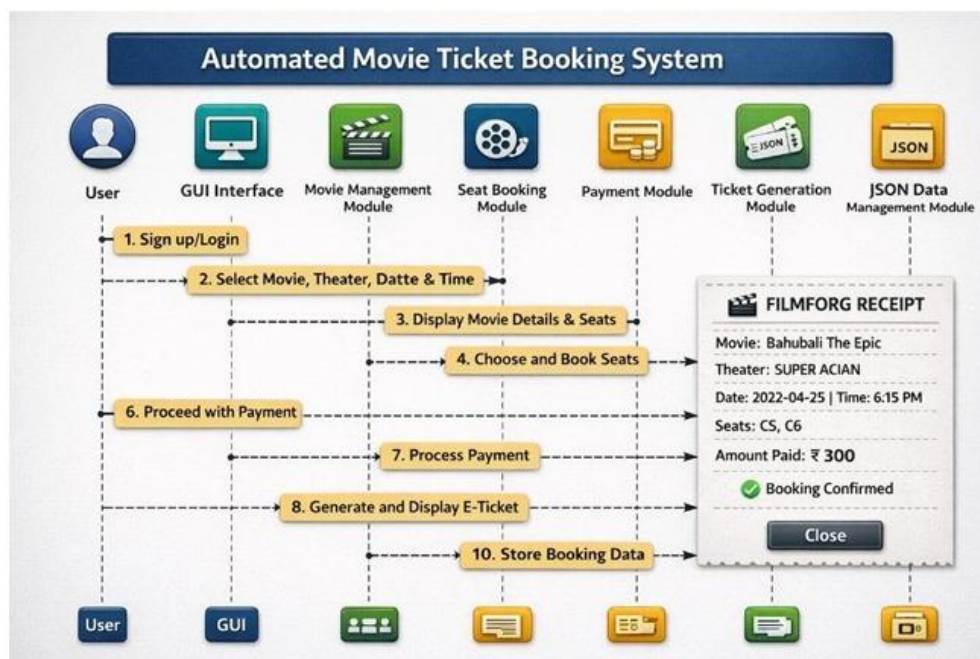


Fig 3.2.3 Sequence Diagram

3.3 MODULE DESCRIPTION

The Automated Movie Ticket Booking System is divided into several modules, where each module performs a specific function in the system.

1.User Interface Module This module allows users to interact with the system. It provides options to view movies, select show timings, choose seats, and confirm bookings through a graphical interface.

2.Movie Management Module This module manages movie details such as movie name, show timings, and available seats. It helps display updated movie information to users.

3.Seat Booking Module This module handles seat selection and checks seat availability. It ensures that a seat cannot be booked more than once.

4.Payment Module This module processes the payment for ticket booking. After payment confirmation, the booking process continues.

5.Ticket Generation Module This module generates the electronic ticket after successful booking and displays the ticket details to the user.

CHAPTER 4 IMPLEMENTATION

4.1 ENVIRONMENTAL SETUP

To run the Automated Movie Ticket Booking System, Python must be installed on the computer. An IDE such as PyCharm or VS Code is used to write and run the program. Required libraries like Tkinter should be available, and project files should be organized properly to execute the application.

4.1.1 Development Environment

The Automated Movie Ticket Booking System is developed using Python as the programming language with Tkinter for the graphical user interface. The development is done in an IDE like PyCharm, which provides tools for coding, debugging, and project management. JSON files are used to store movie and booking data, making the system organized and easy to maintain.

Setup Steps:

Install Python – Download and install the latest version

Install IDE – Use PyCharm or VS Code for development.

Check Libraries – Ensure Tkinter and JSON are available.

Organize Files – Keep Python scripts and JSON data in a project folder.

Run Application – Open the main file in IDE and execute.

Test System – Verify seat selection, payment, and ticket generation.

4.2 IMPLEMENTATION OF MODULES

The Automated Movie Ticket Booking System is implemented using separate modules, each handling specific functionality:

1. User Interface Module

- Developed using Tkinter.
- Provides windows, buttons, and forms for users to view movies, select seats, and book tickets.

2.Movie Management Module

- Manages movie details, show timings, and available seats.
- Retrieves data from JSON files and updates information when required.

3.Seat Booking Module

- Checks seat availability before booking.
- Updates seat status in JSON to prevent double booking.

4.Payment Module

- Simulates payment processing.
- Confirms booking once payment is completed.

5.Ticket Generation Module

- Automatically generates electronic tickets after successful booking.
- Displays ticket details including movie name, seat number, and show time. Each module works independently but communicates with others to complete the booking process efficiently.

4.3 INTEGRATION AND DEVELOPMENT

Development

The development of the Automated Movie Ticket Booking System involves creating a desktop-based application using Python. The process starts with designing the Graphical User Interface (GUI) using Tkinter, allowing users to interact with the system easily. The application logic is implemented to handle movie selection, seat availability checks, booking confirmation, and payment processing. JSON files are used to store movie details, seat status, and booking records, ensuring data is managed efficiently.

Each module, such as Seat Booking, Payment, and Ticket Generation, is developed separately and integrated to work together. Testing is performed at each stage to ensure proper functionality. Finally, the modules are combined to create a fully functional, user friendly system that automates movie ticket booking.

Integration

In the Automated Movie Ticket Booking System, all modules are integrated to work as a complete system. The User Interface module interacts with the Movie Management and Seat Booking modules to display available movies and seats. Once the user selects seats, the Payment module processes the payment, and the Seat Booking module updates the seat status in the JSON database.

After successful payment, the Ticket Generation module creates an electronic ticket and displays it to the user. Integration ensures that all modules communicate smoothly, providing a seamless workflow from selecting a movie to receiving the ticket. This makes the system fully functional and user-friendly.

Testing and Refinement

The Automated Movie Ticket Booking System undergoes rigorous testing to ensure it works efficiently and accurately. Each module, including Seat Booking, Payment, and Ticket Generation, is tested individually to verify its functionality. Afterward, integration testing is performed to ensure that all modules communicate properly, such as updating seat status in the database and generating tickets after payment.

The system is also tested from a user perspective to check usability, navigation, and correctness of booking information. Any errors or bugs identified during testing are fixed, and refinements are made to improve performance, reliability, and user experience.

This ensures that the system is fully functional, user-friendly, and ready for practical use Deployment. Deployment of the system is straightforward. The frontend can be hosted on any static server, while the backend can run on local or cloud-based Python servers. Since the modules communicate through lightweight API calls, the system can scale easily and maintain fast response times. This makes the Hybrid Autocomplete System suitable for real-world applications like search engines, chat applications, and e-commerce platforms where quick and smart suggestions are essential.

CHAPTER 5 EVALUATION

1.1 MODULES

The Automated Movie Ticket Booking System consists of five main modules:

User Interface Module – Evaluated for ease of use and responsiveness

Movie Management Module – Checked for accurate movie and show details.

Seat Booking Module – Ensures correct seat allocation and prevents double booking.

Payment Module – Verified for smooth and accurate payment processing.

Ticket Generation Module – Assessed for timely and correct ticket generation. Each module is evaluated to ensure the system works efficiently and reliably.

Validation Module (AI Frequency Ranking Module)

The Validation Module ensures that all inputs and actions in the Automated Movie Ticket Booking System are correct and reliable. It checks user inputs, such as name, contact details, and seat selection, to prevent errors during booking. The module also validates seat availability before confirming a booking and ensures that payment information is correctly processed. By performing these checks, the validation module prevents mistakes such as double booking of seats, invalid user entries, or incomplete payments, ensuring that the system operates smoothly and provides accurate booking results.

Test Module (Suggestion Generation & Display Module)

The Test Module is used to verify that the Automated Movie Ticket Booking System functions correctly and meets the intended requirements. It checks each module individually and in combination to ensure proper operation.

Key testing areas include:

- **User Interface Testing** – Ensures all buttons, menus, and windows work properly.

- **Seat Booking Testing** – Verifies that seat selection and availability updates are accurate.
- **Payment Testing** – Confirms that payment processing works correctly and triggers booking confirmation.
- **Ticket Generation Testing** – Ensures that electronic tickets are generated with correct details. This module helps identify bugs or errors so they can be fixed, ensuring the system is reliable, accurate, and user-friendly.

5.2 EVALUATION METRICS

The Automated Movie Ticket Booking System is assessed using several evaluation metrics to ensure its efficiency, accuracy, and user-friendliness.

1. **Accuracy** – Checks whether seat booking, payment, and ticket generation are correct without errors.
2. **Response Time** – Measures how quickly the system responds to user actions, such as displaying movies or confirming bookings.
3. **Usability** – Evaluates the ease of use of the graphical interface and clarity of options for users.
4. **Reliability** – Ensures the system functions consistently without crashes or data loss.
5. **Data Integrity** – Assesses whether the system correctly updates and maintains booking records and seat availability.
6. **Security** – Ensures that sensitive data such as payment and user details are handled securely. These metrics help determine the overall effectiveness and performance of the system.

1.2 TEST CASES

1. User Interface Module

- Test Case: Open the application
- Input: Launch the program
- Expected Output: Application opens successfully

2. Movie Management Module

- Test Case: View available movies
- Input: Click on “View Movies”
- Expected Output: List of movies and show timings displayed

3 .Seat Booking Module

- Test Case 1: Select available seat
- Input: Choose a free seat
- Expected Output: Seat marked as booked, no double booking
- Test Case 2: Select already booked seat
- Input: Choose a seat that is already booked
- Expected Output: Error message prompts user to select another seat

3. Payment Module

- Test Case 1: Enter payment details
- Input: Provide card/QR payment information
- • Expected Output: Payment confirmed and booking completed Ticket Generation Module
- Test Case 2: Complete booking
- Input: Confirm booking after payment
- Expected Output: Electronic ticket generated with correct details

4. Validation Module

- Test Case: Enter invalid user data
- Input: Provide wrong or incomplete details
- Expected Output: Error message prompting correct input

5. Data Management Module

- Test Case: Update booking record
- Input: Complete booking process
- Expected Output: JSON file updated correctly with booking details

1.3 RESULTS

The Automated Movie Ticket Booking System performed efficiently during testing. The GUI was responsive, movies and show timings were displayed correctly, and seat booking prevented double bookings. Payments were processed smoothly, and tickets were generated accurately. User input validation and data updates in JSON files worked correctly, making the system reliable, accurate, and user-friendly.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

CONCLUSION:

The Automated Movie Ticket Booking System successfully automates the process of reserving movie tickets, providing a convenient, fast, and reliable solution for both users and theatre management. It eliminates the need for manual booking, reduces errors such as double booking, and ensures accurate record-keeping. The system's modular design, with separate modules for GUI, seat booking, payment, and ticket generation, makes it easy to maintain and scalable for future enhancements. Overall, the project demonstrates how automation improves efficiency and user experience, making movie ticket booking simple and error-free.

Key Achievements:

1. **Automated Ticket Booking** – Successfully implemented a system to book movie tickets without manual intervention.
2. **User-Friendly Interface** – Developed a GUI using Tkinter for easy and interactive user navigation.
3. **Accurate Seat Management** – Ensured correct seat allocation and prevented double booking.
4. **Payment Simulation** – Integrated a payment module to confirm bookings smoothly.
5. **Ticket Generation** – Automated electronic ticket creation with all booking details.
6. **Data Management** – Used JSON files to maintain accurate records of movies, seats, and bookings.
7. **Validation & Reliability** – Implemented input validation to reduce errors and ensure system reliability.
8. **Modular Design** – Designed separate modules for better maintenance and future scalability.

FUTURE ENHANCEMENTS

1. **Online Payment Integration** – Add support for secure online payments using gateways like PayPal, Razorpay, or Stripe.
2. **Web and Mobile Version** – Develop a web-based or mobile application for easier access from anywhere.
3. **Seat Selection Map** – Implement a graphical seat map for users to view and select seats visually.
4. **User Accounts** – Add user registration and login to save booking history and preferences.
5. **Notifications** – Include email or SMS confirmations for booked tickets.
6. **Multiple Theatre Support** – Extend the system to handle multiple theatres and screens.
7. **Analytics Module** – Add reporting and analytics for movie popularity, sales trends, and user behavior.
8. **Cloud Storage** – Use cloud databases to store booking records securely and allow remote access.

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- Python Official Documentation – <https://docs.python.org>
- Tkinter Documentation – <https://docs.python.org/3/library/tkinter.html>
- JSON Module in Python – <https://docs.python.org/3/library/json.html>
- Online resources on automated ticket booking systems and reservation management systems (research papers, tutorials, and academic articles).

APPENDIX AND SAMPLE CODE

A.1 DATA/STRUCTURE DESCRIPTION

The Automated Movie Ticket Booking System uses JSON files to store and manage data.

1. Movie Data – Stores movie name, show timings, and available seats.
2. Booking Data – Stores user details, selected movie, seat number, and payment status.
3. Seat Management – Tracks seat availability to prevent double booking.

Advantages: JSON is easy to read/write, supports nested data, and allows quick retrieval and updates, ensuring smooth operation of booking and ticket generation.

A.2 SYSTEM MODULES / ARCHITECTURE

1. User Interface Module – GUI for viewing movies, selecting seats, and booking tickets.
2. Movie Management Module – Manages movie details, show timings, and seat availability.
3. Seat Booking Module – Handles seat selection and prevents double booking.
4. Payment Module – Processes payments and confirms bookings.
5. Ticket Generation Module – Generates and displays electronic tickets.
6. Validation Module – Checks user inputs, seat availability, and payment details.
7. Data Management Module – Stores and retrieves data using JSON files. This modular architecture ensures the system is efficient, reliable, and easy to maintain.

A.3 TRAINING AND EXECUTION SETUP

1. Development Environment
 - Python is used as the programming language.
 - IDE: PyCharm or VS Code for writing, debugging, and running code.
 - Required libraries like Tkinter (for GUI) and JSON (for data handling) should be installed.
2. Data Preparation
 - Movie details, show timings, and seat availability are stored in JSON files.
 - Test booking records can be preloaded to simulate real-world usage.
3. Execution Steps
 1. Open the IDE and load the main Python script.
 2. Run the script to launch the GUI interface.
 3. Users can select movies, choose seats, and proceed with payment.
 4. The system updates booking records and generates tickets automatically.
5. Testing
 - Test each module individually (seat booking, payment, ticket generation).

- Perform end-to-end execution to ensure smooth workflow and error-free operation. This setup ensures the system is ready for practical use and testing.

A.4 DEPLOYMENT DETAILS

The Automated Movie Ticket Booking System runs on Windows, macOS, or Linux with Python installed. Required libraries include Tkinter and JSON. All Python scripts and JSON files should be in a single project folder. Run the main script to launch the GUI, allowing users to book tickets, make payments, and generate e-tickets. Updates can be made by modifying JSON files or adding new modules.

A.5 RESULTS

The Automated Movie Ticket Booking System was tested and found to be efficient, accurate, and user-friendly. The GUI worked smoothly, movies and show timings displayed correctly, and seat selection prevented double bookings. Payments were processed successfully, and electronic tickets were generated with correct details. User input validation and data updates in JSON files worked as expected, ensuring the system is reliable and error-free for practical use.

A.6 BENEFITS OF THE SYSTEM

1. Time-Saving – Users can book tickets quickly without waiting in long queues.
2. Accuracy – Prevents errors like double booking and ensures correct seat allocation.
3. User-Friendly – GUI interface makes the system easy to use.
3. Efficient Data Management – Stores movies, seats, and booking records in JSON for easy retrieval and updates.
4. Convenience – Users can view movie details, show timings, and book tickets from a single platform.
5. Automated Ticket Generation – Generates electronic tickets instantly after booking.
6. Scalability – New features like online payment or multiple theatre support can be added easily.
8. Reliability – Validation and error checks ensure smooth operation without system crashes.